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Crust Goat Leather Dyeing using Mahogany (*Swietenia mahagoni*) Sawdust Extract and Coloring Time to Get the Best Results

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ABSTRACT

Synthetic dyes used in the leather dyeing process contain allergenic and carcinogenic compounds. Natural dyes sourced from wood industry waste, namely mahogany sawdust can be a substitute for synthetic dyes with the advantages of being more economical, minimizing waste, and relatively safe for the environment. This study aims to determine the effect of the comparison of mahogany sawdust with hot water solvent and the duration of dyeing time on the characteristics of the extract and the best results of dyeing application on crust goat leather, using a Completely Randomized Factorial Design with 2 (two) factors and 3 (three) levels respectively. The first factor is the ratio of mahogany sawdust with solvent during Extraction, 1:5; 1:10 and 1:15 (w/v) and the second factor is the duration of dyeing time, 60, 90 and 120 minutes, each with two repetitions. Data analysis used ANOVA analysis of variance. This research activity resulted in a comparison of mahogany sawdust and water solvent with the best duration of dyeing time in producing colored goat leather products with the criteria of color absorption, rub fastness (wet and dry) and color shading. The results showed the analysis of the highest extract tannin content was 7,66 µg/mL at 1:5 (w/v) extract. The highest color absorption was in the dye extract 1:5 (w/v) and the lowest was in the dye extract 1:15(w/v), both at 60 minutes, 90 minutes, and 120 minutes. All natural dyes from mahogany sawdust showed good color fastness values with a scale above 3/5. The visual result of the dyeing process is reddish brown.

Keywords: goat leather, mahogany sawdust, natural dyes, tannin

INTRODUCTION

Animal skin/hide is a by-product of the livestock slaughtering industry which has the highest economic value. Raw animal skins can be processed into leather products, such as: jackets, shoes, gloves, car seats, belts, hats, and others through a series of process stages. One of the important stages is the finishing stage which aims to give the skin the right texture and structural properties, the quality of touch and color, physicomechanical, and the desired fastness (Hansen et al., 2020). Skin coloring is currently still dominated by synthetic dyes (Sudha et al., 2016). Research has shown that synthetic dyes are thought to release harmful chemicals that are allergenic and carcinogenic and detrimental to human health. The water and air pollution caused by synthetic dyes, especially industrial effluent control, has become a major problem (Khalid et al., 2010).

The use of natural dyes as a substitute for synthetic dyes has been widely used, especially in textile coloring, inks, cosmetics, etc., but the application of natural dyes on the skin is still limited (Khalid et al., 2010). Natural dyes do not contain harmful chemicals, are easy to obtain, easily degraded, and the coloring process also produces non-toxic liquid waste (Gobalakrishnan et al., 2020). Natural dyes produce beautiful and distinctive color effects compared to synthetic dyes, producing products that are exclusive and of high artistic value so that they have an appeal to certain market segments (Rahmawati et al., 2020). Skin dyes ideally meet several criteria, such as cost-effectiveness, flexibility, fastness standards, extent and extent of defects, and reproducibility (Puntener, 2000).

One alternative for the production of natural dyes is to utilize by-products or waste from wood and the agricultural industry. Natural dyes extracted from agricultural/non-agricultural waste and by-products are one of the potential strategies to address the increasing consumer demand and lead to the minimization of waste/pollution levels (Rather et al., 2020). Mahogany (*Swietenia mahagoni*) is one type of wood that is widely available in Indonesia and contains high-quality dyes. Mahogany wood is currently widely used as a furniture material because the planting period is quite short, the price is not too expensive, the texture is softer than other woods so it is easy to shape, has straight wood fibers similar to teak so it is widely used as a substitute for teak wood. Mahogany sawdust is waste from the furniture industry made from mahogany (Swietenia mahagoni) in the form of red brown coarse powder. This mahogany wood powder is abundantly available and has not been utilized optimally, its use is still limited as a solid fuel and a small part is used as a mixture in the manufacture of particleboard. Mahogany sawdust extract has been tested as a natural colorant for non-conventional skin (lizard skin, snake skin, snapper skin and chicken feet) producing a brown color (Kasmudjiastuti, 2017). Brown, black, and grey are the most popular colors for shoe products, upholstery, and garments (Puntener, 2000).

Natural dyes that have been applied to the skin include Bixa Orellana as a natural dye for chrome-tanned sheepskin, good wet and dry rubbing resistance values, namely a scale of 4/5 (Selvi et al., 2013), mangrove bark extract as a fish leather dye. Tilapia with different concentrations and soaking time had a significant effect on tensile strength, tear strength, elongation, organoleptic test, and wet and dry paint rubbing resistance test (Ramadhan et al., 2016), extract of Hambo-hambo dye (Cassia singueana) on sheepskin chrome tanned with copper sulfate mordant, aloe vera, and mango peel produced the best K/S value (darkness of a color) in samples by staining at a temperature of 70ºC, time of 60 minutes and pH 4 in samples with natural mordant (Berhanu & Ratnapandian, 2017) . Crust goat leather dyeing using natural dyes from mahogany sawdust extract and alum mordant has never been done.

Mahogany wood has a fairly high economic value and is not easy to change (BPHWLT, 2017). Mahogany wood contains flavonoids, tannins, and quinones which are components of dyes (Febriana et al., 2016). Tannins can be extracted with water or water mixed with other solvents such as methanol, ethanol, acetone, NaOH. Extraction of tannins with NaOH solvent has been carried out by Guo et al. (2020), tannin extraction from Coriaria nepalensis stems with variations in extraction time, extraction temperature, solvent-ingredient ratio, and NaOH concentration resulted in a yield of 16.02±0.08% at room temperature. Extraction at 82ºC for 63 minutes with 0.22% NaOH at a ratio of 23 mL g-1(v/w). The tannin extraction process with hot water will give better results (Das et al., 2020). Particle size affects the extraction process. The ratio of solids to solvents and extraction temperature are very important for tannin extraction (Arina & Harisun, 2019). Utilization of mahogany bark as a dye for silk batik with variations in pH (2, 6, and 12) showed good color fastness (scale 4-5), the highest flavonoid and tannin content in extracts with alkaline pH (Lestari et al., 2020). Parameters of temperature, staining time and pH affect the results of natural coloring (İşmal & Yildirim, 2019). Mahogany wood extract as a red snapper skin colorant produces optimal physical properties of the skin including tensile strength, elongation, and paint rubbing resistance (wet and dry) (Kasmudjiastuti & Widhiati, 2002).

In this study, goat skin was chosen because goat skin is the most economical leather compared to other conventional leather. Although the price is affordable, goatskin is also strong and durable, the texture is harder than cowhide but also supple, lightweight and water-resistant. Can be used for making gloves, book covers, shoes, bags, clothes. This study aims to see the effect of the comparison of mahogany sawdust with water solvent and the duration of dyeing on the characteristics of the extract and the results of the application of dyeing on the crust goat leathers. After that, it was continued with decision making using the Exponential Comparison Method (MPE). This method will produce the best combination of the ratio of materials to solvents and the duration for dyeing in producing colored goat leather products.

MATERIAL AND METHOD

Materials and Equipment.

The raw material used in this research is pickel goat skin which is obtained from one of the leather tanning industries in Yogyakarta. Mahogany sawdust (Swietenia mahagoni) was obtained from Dusun Seropan, Dlingo, Bantul. Chemicals for the skin coloring in soda ash, teepol, sincal ms, alum (Al2(SO4)3.K2SO4.24H2O) were obtained from Karyanti Stores, Yogyakarta. The equipment used for the skin dyeing process includes a set of coloring drums, scales, an electric stove, thermometer, pH paper, glassware, and test equipment, including a crock meter (AATCC model M238 AA), UV-Vis Spectrophotometer Shimadzu UV-1601PC, Konika chromameter Minolta CR-400.

Research Design.

This study was conducted to examine the effect of comparison of mahogany sawdust with water solvent and the duration time of dyeing on the characteristics of the extract and the results of dyeing on the crust goat leather. The experimental design used was a completely randomized factorial design (CRD) with 2 (two) factors and 3 (three) levels each. The first factor is the ratio of mahogany sawdust with solvent during Extraction, 1:5; 1:10 and 1:15 (w/v) and the second factor is the length of dyeing time, 60, 90 and 120 minutes, each with two repetitions. Data analysis used ANOVA analysis of variance and Duncan's follow-up test. The research treatment is presented in Table 1.

Table 1. Research Treatment

No	The ratio of mahogany sawdust and solvent (w/v)	Dyeing time (minutes)
1	1:5	60
2	1:10	90
3	1:15	120

Procedures

Raw Material Preparation

Goat skin used is goat skin that has undergone a pre-tanning process (pretanned), including soaking, liming, hair removal, deliming, and others aimed at removing hair, blood, preservatives, and dirt from the skin. The pretanned skins were tanned with chromium trioxide (Cr^2O^3) then the skins that had been tanned were posttanned (shaving, neutralization, and oiling) to obtain kras goat skins that were ready to be colored. Selected goat skin measuring 0.6-0.7 m^2, thickness 0.6 mm, every 1 treatment used a quarter of the skin.

Extraction of dyestuffs from mahogany sawdust

The process of making a dye solution from waste mahogany sawdust follows the method used by Kasmudjiastuti & Widhiati (2002). Mahogany sawdust dried in the sun for 12 hours. Dry sawdust is added with water in a ratio of 1:5; 1:10 and 1:15 (w/v). A total of 1000 mL of the solution was boiled until it boils and then heated for 30 minutes at 100ºC until the solution remaining half of the volume of the initial solution, then filtered with a filter cloth to separate the powder from the dye liquid. The resulting filtrate is called Solution I. The remaining mahogany sawdust is added again with water according to the initial ratio twice, heated in the same way as Solution I to obtain Solution II and Solution III.

Dye Characterization

Natural dyes from mahogany sawdust were characterized, including pH and tannin content. The pH of the dye solution was measured using pH paper, while the tannin content was quantitatively measured using a UV-Vis Spectrophotometer (Lestari et al., 2020).

Dyeing Process

The dye from mahogany sawdust with the ratio of materials: solvent is 1:5; 1:10; and 1:15 (w/v) were applied to the crust goat leather with variations in total dyeing time of 60, 90, and 120 minutes. The mordant used was alum ($Al_2(SO_4)_3.K_2SO_4.24H_2O$) with the pre-mordanting method. Alum used each as much as 3% of the weight of the leather. The rotating speed of the coloring drum is 15 rpm. The complete coloring process is presented in Table 2.

The process of coloring the crust goat leather was carried out without the addition of synthetic dyes. The coloring process is carried out in 3 (three) stages, each starting with the most dilute solution (solution III), solution II, and finally the most concentrated solution (solution I). The dyeing mechanism was carried out at pH 5-6 according to the method in the vegetable tanning process from wood extract. Dyeing begins with a more dilute solution because the aqueous solution easily penetrates the skin fibers due to the smaller particle size compared to the concentrated solution (Kasmudjiastuti, 2006). At the beginning of the dyeing process, the temperature of the dye added is 40ºC as much as half of the amount of dye that will be used then after half the time, the dye is added with a temperature of 60ºC.

Tabel 2. Leather Dyeing Process

Process	Material	Amount (%)		Time (minutes)
Wetting back /	Water	300	*	
Neutralization	Soda ash	3	*	60
	Teepol	3	*	*Drain*
Mordanting	Water	100	#	
	Alum	3	#	45
				Drain
Dyeing	Solution III @	150	#	X, *drain*
	Solution II @	150	#	Y, *drain*
	Solution I @	150	#	Z, *drain*
Fatliquoring	Water 40^{0}C	100	#	
	Oil	6	#	60
				Drain
Aging				

Note: * = dry weight of crust goat leather, # = wet weight of crust goat leather; @ = natural dye extracted with water solvent 1:5; 1:10; 1:15 (w/v); X = dyeing time 20, 30, 40 minutes; Y = dyeing time 20, 30, 40 minutes; Z = dyeing time 20, 30, 40 minutes; The dyeing time of 60, 90, and 120 minutes was divided into three steps: the first step used Solution III, the second used Solution II and the third used Solution I; Dyeing time 60 minutes = Solution III, II, and I each for 20 minutes;Dyeing time 90 minutes = Solutions III, II, and I each for 30 minutes; Dyeing time 120 minutes = Solutions III, II, and I each for 40 minutes

Product Characterization

Quantitative Analysis of Tannins

The determination of total tannin content uses tannic acid as a standard solution (Lestari et al., 2020). The analysis step begins with determining the maximum wavelength by measuring the concentration of tannic acid solution of 50µg/ml in the wavelength range of 400-800 nm with a UV-Vis Spectrophotometer. Measure the standard solution by making the concentration of tannic acid 0µg/mL, 1µg/mL, 2µg/mL, 3µg/mL, 4µg/mL, 6µg/mL, 10µg/mL pipetted from a standard solution of tannic acid with a concentration of 1000 g/mL, then 0.4 mL of Folin Ciocalteau reagent was added, shaken and left for 4-8 minutes. Added 4.0 mL of 1% NaOH solution, shaken until homogeneous, and added aquabidestilata up to 100 mL, measured at the maximum wavelength, made a calibration curve of the relationship between the concentration of tannic acid and absorbance (Sumadewi & Puspaningrum, 2021).

The tannin content of natural dyes was determined by weighing 10 mL of dye taken from solution I + solution II + solution III from each dye extracted 1:5; 1:10; 1:15 (w/v) (dyes/solvents). The dye extract was dissolved with 10 mL of aquabidestilata, homogenized, pipette 1 mL of the solution, then added with 0.4 mL of Folin ciocalteau reagent, shaken, and allowed to stand for 4-8 minutes then added 4.0 mL of 1% NaOH, shaken until homogeneous, added aquabidestilata up to 50 ml. Take 25 mL and fill it with aquabidestilata to a volume of 50 mL and then measure the absorbance at a wavelength of 653 nm.

Color Absorption Test

Measurement of color absorption according to the method of Mahdi et al. (2021). Where A_0 is the absorbance of the initial dye solution, and A1 is the absorbance of the dye solution after the dyeing process at max. The absorbance was measured using a Shimadzu UV-Vis Spectrophotometer UV-1601PC.

$$\%\, Color\ absorption\ test = \frac{(A_0 - A_1)}{A_0} x\ 100$$

Color Resistance

The color fastness test stage uses a Crockmeter to perform a wet-dry rub test. Grey scale standards (Grey scale) to assess color changes in the color fastness test, and staining scale standards (Staining scale) to assess color staining on white fabrics used in color fastness tests. The value of the grey scale determines the level of change or contrast of skin color before and after rubbing, the level of values from the best to the lowest values are: 5 ; 4/5; 4; 3/4; 3; 2/3; 2 ; 1/2; 1. Tests are carried out according to ISO 105-A02:1993(E). The value of the staining scale determines the level of staining of white cloth before and after rubbing, the level of values from the best to the lowest values are: 5 ; 4/5; 4; 3/4; 3; 2/3; 2 ; 1/2; 1. Tests are carried out according to ISO 105-A03:1993. According to SNI 4593:2011, the requirements for the grey scale value of the cover paint for dry rub resistance are at least 4.5, and the grey scale value for wet rub resistance is at least 4 (BSN, 2011)

Color Difference Test

Differences in skin color were analyzed using the Konika Minolta CR-400 Chromameter. Color measurements were compiled using the Commission Internationale de Eclairage (CIE) color measurement system with 100 standard observer data (Ramalingam & Jonnalagadda, 2017). The values L*, a*, and b* are variables in the CIELAB color space. A more negative L* value indicates a darker color, and a more positive L* value for a lighter color. Negative a* values imply more green, and more positive a* values for more red. More negative b* values mean more blue, and more positive b* values for more yellow (Dave et al., 2016; (Ramalingam & Jonnalagadda, 2017).

RESULT AND DISCUSSION

Dye Characterization.

The results of the pH test and tannin content of the natural dye extract of mahogany wood powder are presented in Table 3. The pH value of mahogany sawdust extract is 5.5 so it is included in the acid dye group. According to Lestari et al. (2020), the use of mahogany wood extract for dyeing silk batik cloth under acidic (pH 2), neutral (pH 6), and alkaline (pH 12) conditions had no significant effect on the quality of dyeing, all three resulted in good coloring quality (scale 4-5).

The highest tannin content in the dye extract was 1:5 (w/v), followed by 1:10(w/v), and the lowest was 1:15(w/v). Differences in solvent ratio resulted in significantly different extract tannin levels. This mahogany sawdust extraction is solid-liquid Extraction, using a liquid solvent to separate one or more constituents in the solid mahogany sawdust. At the end of the extraction process, several solutions containing solvent and extract were obtained, separated from the solid particles due to the adhesive force. The solution taken from the solid has the same concentration of the active compound as the extract. At equilibrium, it is assumed that the total amount of the active compound is the amount dissolved in the solvent (Kristijarti & Arlene, 2012). The dye extract used in this study is still a mixture of solute (dye) and water as a solvent. The smaller the number of solvents with the same amount of solids, the more concentrated dye extract will be produced so that the 1:5 w/v solids: solvent extract has the higher tannin content followed by 1:10 w/v and the smallest 1:15 w/v.

This corresponds to the density of the resulting color. The smaller the material: solvent ratio, the higher the extracted tannins. The total tannin content describes the amount of dissolved dye in the mahogany wood extract used to color goat leather (Lestari et al., 2020). Among several methods, the tannin extraction method using hot water is still used in the industry (Kemppainen et al., 2014) and in the laboratory (Tascioglu et al., 2013). This is because the method is simple and low-cost. In addition, some researchers even reported that the number of condensed tannins and hydrolyzable tannins was highest when the solvent was hot water (Das et al., 2020).

Table 3. Result of pH test and tannin content of dye extract

No	Extract solution (w/v)	Color	pH	Tannin Content (µg/mL)
1	1:5	brown	$5,5^a$	$7,66^a$
2	1:10	brown	$5,5^a$	$6,09^b$
3	1:15	brown	$5,5^a$	$4,66^c$

Note: Data is the average result of two replications; Data followed by different letter marks in the same column indicate a significant difference (P < 0.05)

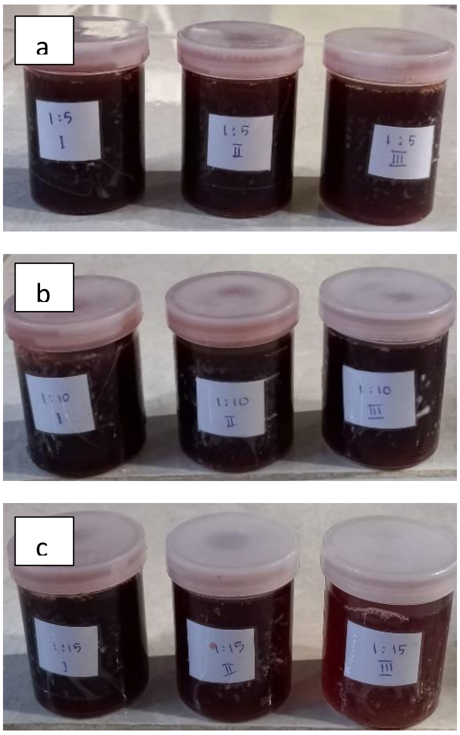


Figure 1. Mahogany sawdust extract (a) 1:5 (w/v) (b) 1:10(w/v) (c) 1:15 (w/v)

Color Absorption

The color absorption value was tested by looking at the initial dye solution's absorbance value and the dye solution's absorbance value after the dyeing process at a maximum wavelength of

443.80 nm. The value of color absorption is presented in Figure 2.

The highest color absorption was 1:5 (w/v), followed by 1:10(w/v) and 1:15(w/v), both at 60 minutes, 90 minutes, and 120 minutes. The color absorption of the 1:5 (w/v) dye extract significantly differed from that of the 1:15(w/v) dye extract. This is because the tannin content of the 1:5(w/v) extract is higher than the 1:10(w/v) and 1:15(w/v) dye extracts. Tannins are complex polyphenolic compounds. Tannins contain phenolic compounds that can form hydrogen bonds with the carboxyl groups of goat skin protein fibers. Furthermore, there are two other possibilities involved; (a) the anionic charge on the phenolic group forms an ionic bond with the cationic (amino group) on the goat skin protein substrate; and (b) covalent bonds can also be formed through the interaction between quinine or semiquinone groups present in tannins and reactive groups in skin protein fibers (Punrattanasin et al., 2013). Differences in tannin levels in plants will affect the yield and quality of the dye produced (Sofyan et al., 2018). The higher the tannin content, the higher the content of phenolic compounds and a lot of protein bound to tannins, so the color absorption is higher.

Color absorption at the dyeing time of 60 minutes, 90 minutes, and 120 minutes did not show a significant difference for the same type of extract. This can be caused by the level of color absorption into the skin has reached equilibrium (skin collagen fibers have been saturated) resulting in the dye only sticking to the surface of the skin (Suparno et al., 2016).

Color Resistance

The results of the color fastness test are presented in Table 4. Comparison of materials with solvents did not affect the value of color fastness, both on fabric staining and skin discoloration. Fabric stains and skin discoloration after dry rubbing tend to show a larger greyscale scale than after wet rubbing. Natural dyes from mahogany sawdust showed good color fastness values with a scale above 3/5. This proves that the dye extract from mahogany wood powder can be used as a dye for kras goat skin.

The pre-mordanting process using alum ($Al_2(SO_4)_3.K_2SO_4.24H_2O$) has formed a good complex between metals, dyes, and fibers (İşmal & Yildirim, 2019). Color fastness properties are related to good dye fixation with fiber, better dye penetration, and chemical reaction rate between dye and fiber (İşmal et al., 2016). The chemical reaction between dyes and fibers is the formation of hydrogen bonds between the carboxyl groups of leather protein fibers and phenolic compounds from natural dye components. Furthermore, an ionic bond is formed between the skin protein's amino group (cationic) and the anionic charge on the phenolic group of natural dyes (Lestari et al., 2020).

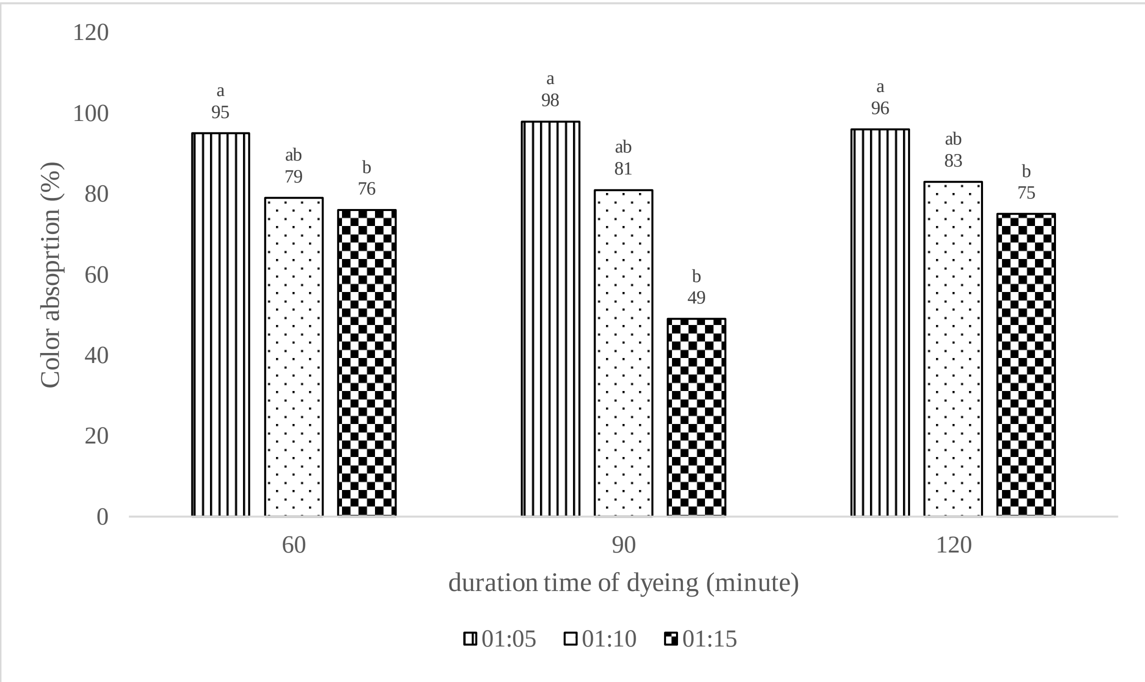


Figure 2. Color absorption chart

Table 4. Colorfastness test results

Treatment	Greyscale			
	Fabric Staining		Skin Discoloration	
	Wet	Dry	Wet	Dry
1:5 60 menit	4	4	4/5	5
1:5 90 menit	4	4	4/5	4/5
1:5 120 menit	4/5	4/5	4/5	5
1:10 60 menit	3/5	4	4	4
1:10 90 menit	4	4	4	4
1:10 120 menit	4	4	4/5	4/5
1:15 60 menit	4	4	4/5	5
1:15 90 menit	4	4	4/5	5
1:15 120 menit	4	4	4/5	4/5

Color Difference

The L* value of colored goat skin samples in this study ranged from 57.49-70.34, as shown in Figure 3. The lower the dye concentration with the same dyeing time, the higher the L value because the lighter the color. This is to the research of Suparno et al. (2016). The dye concentration of 3% resulted in the highest L* value compared to the dye concentration of 6% and 9% at the same dyeing time.

The * value of colored goat skin samples was 7.84-14.86, as shown in Figure 4. At 60 minutes of dyeing, the lower the dye concentration, the lower the * value. The higher the concentration of dye used, the higher the intensity of the red color. Meanwhile, at the dyeing time of 90 and 120 minutes, the dye concentration (comparison of material: solvent 1:10 w/v) showed the highest a* value.

Based on the measurement results, the b* value obtained is positive. The smallest b* value is 23.27, and the largest is 33.85. A high b* value indicates a higher intensity of the yellow color. The value of b* at 60 minutes of dyeing decreased as the dye concentration decreased or was highest at a concentration of 1:5 w/v (material: solvent ratio). The value of b* at the dyeing time of 90 minutes and 120 minutes was highest at a concentration (comparison of (material: solvent) 1:10 w/v. The value of b* tends to decrease as the dye concentration decreases. This result is different from previous research. The concentration of natural dyes in the dye solution affects the brightness value of the dyed skin. High brightness values are observed at low dye concentrations. Samples dyed with low concentrations of Adhatoda vasica extract shift towards the yellow region of the red-yellow coordinates of the CIELab color space with light, bright colors, and less full shades. (Rather et al., 2016).

At a dye concentration of 1:5 (w/v), the highest b* value was at 60 minutes of dyeing, followed by 120 minutes and the lowest at 90 minutes of dyeing. This shows that the dyeing time of 60 minutes has not reached equilibrium, so the color is still not dark (lots of yellow), and equilibrium is reached at the dyeing time of 90 minutes. At a dye concentration of 1:10 (w/v), the lowest b* value was at 120 minutes of dyeing. The b* value at 1:15 (w/v) dye concentration was highest at 120 minutes, followed by 90 minutes, and the lowest b* value at 60 minutes. The lower the dye concentration, the shorter it takes to reach equilibrium. Color test results are presented in Table 5.

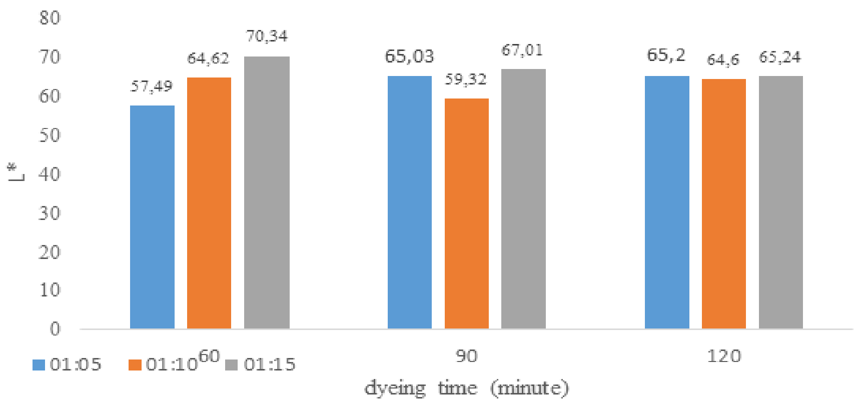


Figure 3. The relationship between dye concentration (material: solvent ratio) and the L* value of goat skin at various dyeing times

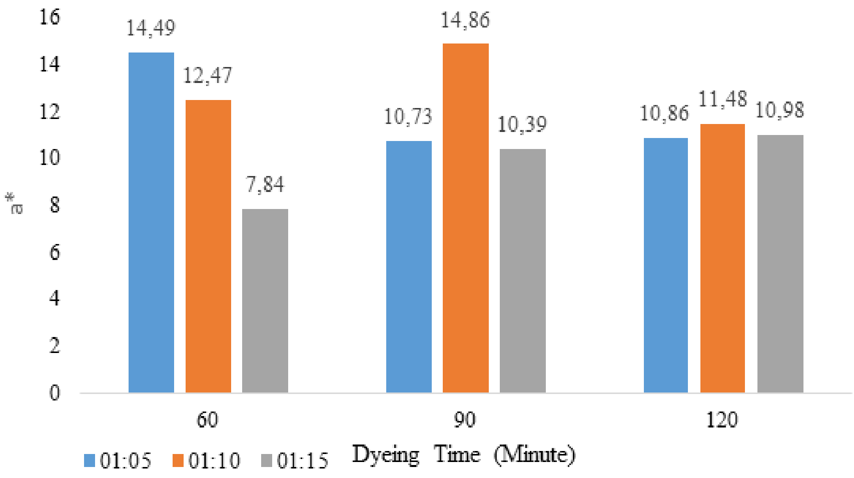


Figure 4. Relationship between dye concentration (material: solvent ratio) and goat skin a* value at various dyeing times

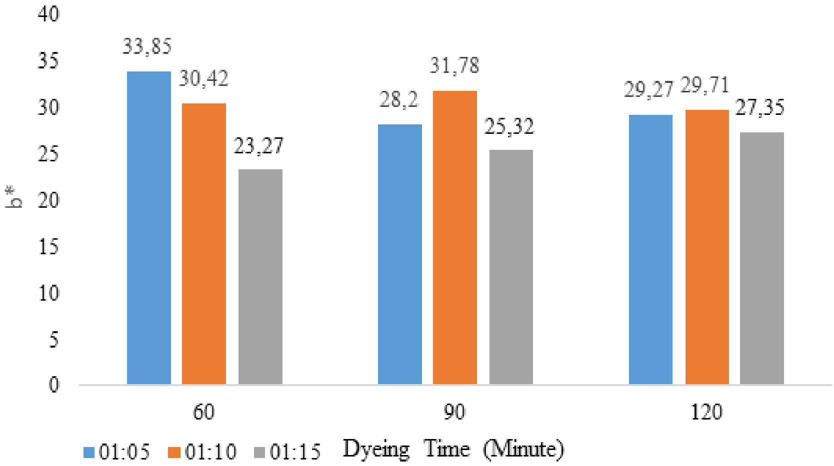
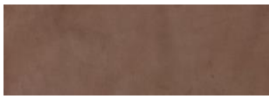


Figure 5. Relationship between dye concentration (material: solvent ratio) and goat skin b* value at various dyeing times

Table 5. The result of coloring goat leather

Treatment	Color	Treatment	Color
1:5 b/v, 60 menit		1:10 b/v, 120 menit	
1:5 b/v, 90 menit		1:15 b/v, 60 menit	
1:5 b/v, 120 menit		1:15 b/v, 90 menit	
1:10 b/v, 60 menit		1:15 b/v, 120 menit	
1:10 b/v, 90 menit			

CONCLUSION

The concentration of dye (amount of material: solvent) significantly affected the tannin content and color absorption. The highest tannin content was 1:5 (w/v) extract with a 7.66 g/mL value. The highest color absorption was 98% at 1:5 (w/v) dye extract with 90 minutes of staining time. Dye concentration and dyeing time did not significantly affect the color fastness. Colorfastness is a good value above 3/5. The average value is 4 - 4/5 (slight fade). The visual result of the dyeing process is reddish brown.

CONFLICT OF INTEREST

No conflict of interest exists with any party related to the materials discussed in the paper.

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